Jeffrey Blay

Greensboro, NC | (240) 756-1706 | jeffreyblay7@gmail.com | LinkedIn | Portfolio | GitHub

Professional Summary: Geospatial data scientist and spatial data engineer with 5+ years of experience building remote-sensing and machine-learning pipelines over large-scale geospatial datasets. Specializes in urban-environment modeling, ETL pipelines for satellite image analysis, and end-to-end spatial AI systems across Python, Google Earth Engine, and the ArcGIS/QGIS ecosystems.

Professional Experience

Geospatial Research Engineer **Jun 2026 – Present**
Remote Sensing & GIS Lab, NC A&T State University

- Building a QGIS plugin that deploys CNN and transformer-based flood-depth models for on-demand inference by end users.
- Assessing disaster impacts on crop-specific land cover and vulnerable communities using federal datasets and Google Earth Engine.

Graduate Research Assistant - NASA Flood Project **Sep 2025 – May 2026**
Remote Sensing & GIS Lab, NC A&T State University

- Built and optimized transformer segmentation architectures (Swin-UNet, SegFormer) in PyTorch for urban flood-depth prediction from geospatial datasets.
- Developed a physics-informed AI framework with terrain-aware hydrostatic constraints, improving the physical consistency of flood modeling.
- Benchmarked physics-informed against conventional deep-learning models across four architectures, reducing surface-gradient violations by ~53%.

Geospatial Data Science Research Assistant **Jun – Aug 2025**
NASA DEAP, NC A&T State University

- Developed and evaluated CNN models (UNet, Attention UNet, UNet++) for flood-depth modeling using geospatial datasets.
- Published ezprocess, an open-source geospatial preprocessing library (PyPI/GitHub) that cut data-prep time by ~70% across ML workflows.

Graduate Research Assistant **Sep 2024 – May 2025**
Remote Sensing & GIS Lab, NC A&T State University

- Built geospatial ETL pipelines using python (rasterio, GDAL) ingest, align, and preprocess ~347 million pixel-level records from multi-source raster data for training computer vision models.
- Maintained codebases, experiments, and version control with Git/GitHub for collaborative development across research teams.
- Led a 3-person team to design and publish a novel flood-depth benchmark dataset for reproducible flood inundation modeling.

Geospatial Data Science Research Assistant - NSF Multimodal Data Fusion **Jun – Aug 2024**
GEMS Institute, NC A&T State University

- Built ETL workflows using NumPy and scikit-learn to prepare 700k+ geospatial records for Random Forest and XGBoost flood prediction models.
- Built python web-scraping workflows leveraging Beautiful Soup and SQLite to collect and curate custom image classification datasets for machine learning applications.

Education

Ph.D., Applied Science & Technology - Geospatial Data Science **Expected Dec 2026**
North Carolina A&T State University

M.E.S., Environmental Science – Urbanization Science **2023**
Yale School of the Environment

Certifications : Foundations of AI Engineering (CodePath) **2026**

Technical Skills

Geospatial & Remote Sensing: ArcGIS Suite, QGIS, Google Earth Engine, ENVI, SNAP
Programming: Python, R, SQL
Data & Cloud: ETL pipelines, Apache Airflow, PostgreSQL, SQLite, PostGIS, Docker, CI/CD, Git/GitHub, HPC, FastAPI